



TSHELO

Technical Architecture

Bank of Botswana Regulatory Sandbox Application

COMPANY

Data Sentinels

PRODUCT

Tshelo

SUBMITTED

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CONTACT

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PRESENTATION TIER

React Native (Expo) mobile app for iOS and Android

APPLICATION TIER

6 core modules: Auth, Fund, Contribution, Expense, Report, Notification

DATA TIER

Supabase PostgreSQL, Auth, Storage, AML Log, Backup Archive on AWS

EXTERNAL INTEGRATIONS

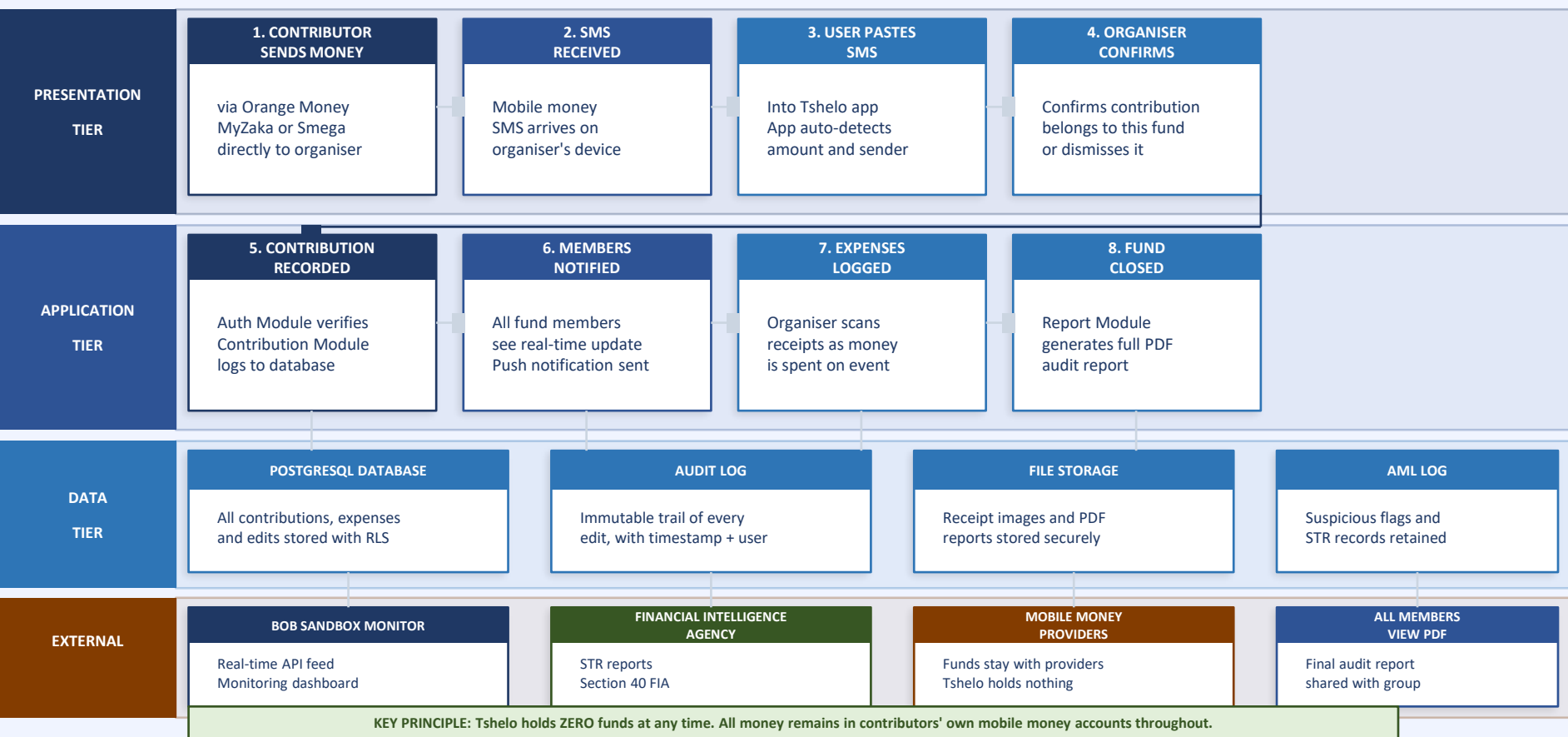
Orange Money, MyZaka, Smega, Google Vision API, FIA, NBFIRA, BoB Sandbox

KEY PRINCIPLES

- Zero fund custody
- AES-256 encryption at rest
- TLS 1.2+ in transit
- Row Level Security
- OTP authentication
- 99.9% uptime SLA
- 24hr auto backups
- Invite-only closed loop
- Immutable audit trail

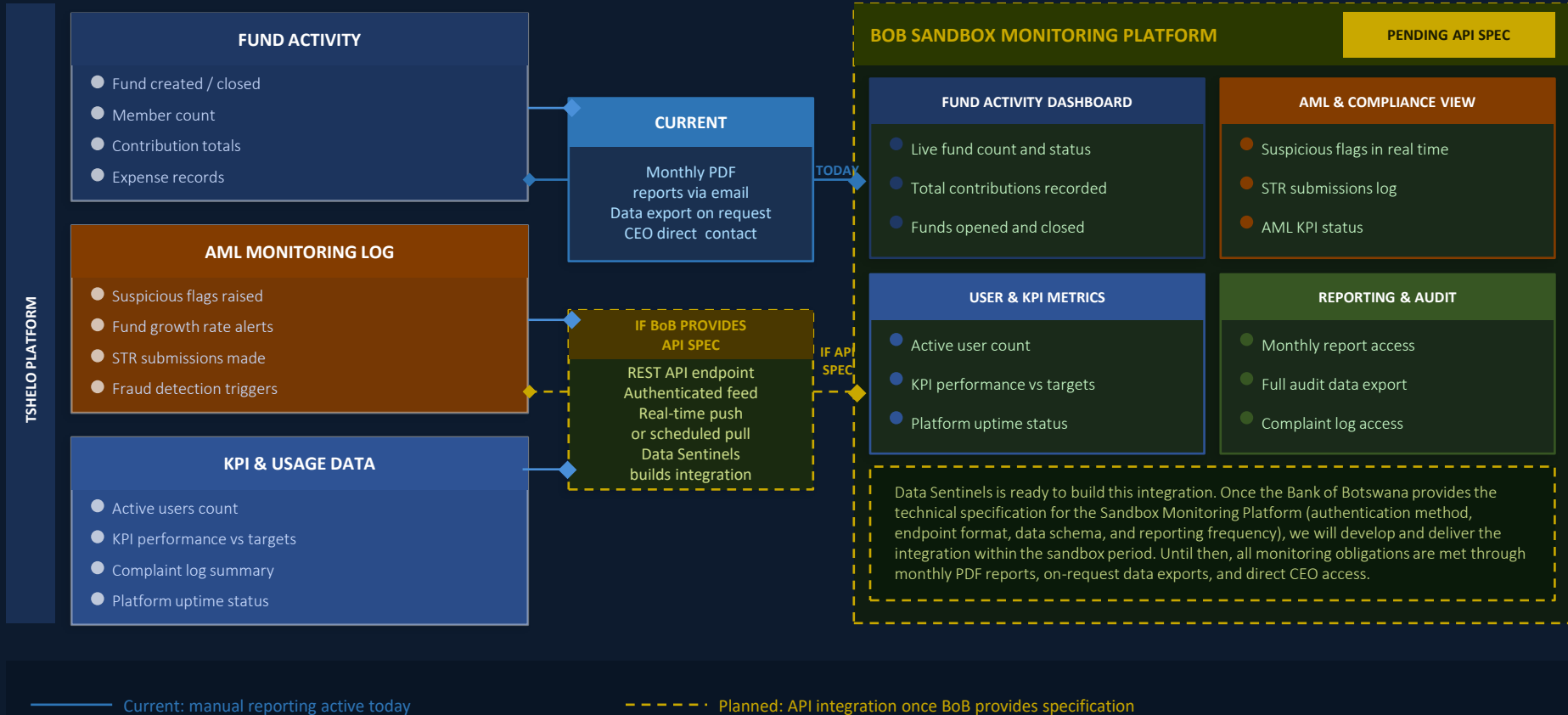
High-Level Process Flow

Presentation > Application > Data > External



BoB Sandbox Monitoring Integration

Architecture | Data Flow | Pending BoB API Specification



TSHELO MOBILE APP

React Native (Expo)
Cross-platform iOS & Android

Fund Dashboard (real-time)

Contribution Recording

SMS Payment Detection

Receipt Scanning (OCR)

PDF Audit Report Export

Member Invitations

Push Notifications

Offline PDF Access

ANDROID

Google Play Store
APK direct install (fallback)

iOS

Apple App Store
TestFlight (sandbox testing)

FUND ORGANISER

Creates fund
Records contributions
Logs expenses
Generates reports

FUND MEMBER

Views contributions
Sees expenses
Flags disputes
Downloads PDF

BOB SANDBOX MONITOR

Real-time API feed
(planned)
Monitoring dashboard
sandbox@bob.bw
Bank of Botswana

AUTH MODULE

- OTP Verification
- Session Management
- Phone Registration
- Account Security

FUND MODULE

- Create & Close Funds
- Fund Goal Setting
- Member Invitations
- Fund Type Selection

CONTRIBUTION MODULE

- SMS Auto-Detection
- Manual Entry
- Contribution Confirm
- Immutable Audit Trail

EXPENSE MODULE

- Receipt Photo Capture
- AI OCR Extraction
- Expense Logging
- Balance Calculation

REPORT MODULE

- PDF Generation
- Progress Charts
- Full Audit Export
- WhatsApp Sharing

NOTIFICATION MODULE

- Push Alerts
- Contribution Reminders
- Fund Updates
- SMS Invites

POSTGRESQL DATABASE	AUTH STORE	FILE STORAGE	AML MONITOR LOG	BACKUP ARCHIVE
<i>Supabase (AWS)</i> - User Profiles - Fund Records - Contribution Data - Expense Records - Edit History Row Level Security on all tables	<i>Supabase Auth</i> - OTP Tokens - Session Tokens - Phone Verification - Auth Logs Session expiry OTP only	<i>Supabase Storage</i> - Receipt Images - PDF Reports - Fund Assets Access controlled per fund member	<i>Internal secure store</i> - Suspicious Flags - Fund Growth Log - STR Records - FIA Reports 5-year retention FIA accessible	<i>AWS S3</i> 24hr Auto Backup 5yr Retention - Point-in-time Recovery Encrypted archive Disaster recovery
INFRASTRUCTURE: Supabase Cloud AWS-backed AES-256 Encryption at Rest TLS 1.2+ in Transit Row Level Security 99.9% Uptime SLA 24-hour Automated Backups				

External Integrations

Mobile Money | APIs | Regulatory Bodies

ORANGE MONEY

Mobile Money Provider

- SMS notification reading
- Device-level only
- User permission required
- BoB Licensed

ACTIVE (SMS)

MYZAKA (MASCOM)

Mobile Money Provider

- SMS notification reading
- Device-level only
- User permission required
- BoB Licensed

ACTIVE (SMS)

SMEGA (BTC)

Mobile Money Provider

- SMS notification reading
- API integration
- Discussions in progress
- BoB Licensed

API IN DISCUSSIONS

GOOGLE VISION API

Receipt OCR Engine

- Receipt text extraction
- Vendor identification
- Amount detection
- Planned v2.0

PLANNED v2.0

FINANCIAL INTELLIGENCE AGENCY

Regulatory Body

- STR reporting channel
- Section 40 FIA
- AML compliance
- Record access

POLICY ACTIVE

BANK OF BOTSWANA

Sandbox Regulator

- Sandbox monitoring API
- Monthly reporting
- Regulatory oversight
- sandbox@bob.bw

SANDBOX ACTIVE

Technology Stack Summary

MOBILE APP

React Native (Expo)

Cross-platform iOS and Android. Distributed via Google Play and App Store.

BACKEND

Supabase (PostgreSQL)

Database, Auth, Storage. Row Level Security. AWS-backed. 99.9% SLA.

AUTHENTICATION

Supabase Auth (OTP)

Phone number and OTP. No passwords. Session token expiry on inactivity.

ENCRYPTION

AES-256 + TLS 1.2+

All data encrypted at rest and in transit. No financial credentials stored.

SMS DETECTION

Device-level (Android)

Parsed locally on device with user permission. Raw SMS never sent to server.

OCR

Google Vision API

Receipt text extraction. Vendor and amount identification. Planned v2.0.

HOSTING

Supabase Cloud (AWS)

SOC 2 Type II certified. 24hr automated backups. 5-year record retention.

AML MONITORING

Internal + FIA reporting

Fund growth monitoring. STR reporting per FIA Section 40. CEO designated officer.